BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. −1 or 101.

Q2. Passes a function's result back to the caller.

Q3. 3.

Q4. Index 0 does not exist, causing an out-of-range error.

### You Do pages · Independent questions

Q1. [4] Starts best=2 [1]; 5 greater so best=5 [1]; 1 ignored [1]; returns 5 [1].

Q2. [6] Set smallest=values[0] [1]; loop over remaining items [1]; if item<smallest [1] then smallest=item [1]; return smallest [1]; array must be nonempty [1].

Q3. [6] Original best=−4 [1]; update to −2 [1]; ignores −9 [1]; returns −2 [1]; initialising to zero would incorrectly return 0 [1] even though zero is absent [1].

### Exit Ticket · Model answer

False. Arrays containing only negative values would produce an incorrect result.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding



RHS COMPUTING / OCR H446 / mixed LESSON 236 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# LESSON 236 · Writing, following and documenting algorithms

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Give a test for a range 0–100 that should be rejected.

Q2. What does return do?

Q3. What is the last index for an array of length 4?

Q4. Why is a[0] unsafe for an empty array?

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

algorithm — Precise finite steps that solve a problem.

precondition — Requirement inputs must meet before execution.

postcondition — Condition guaranteed if the algorithm completes correctly.

invariant — Property maintained during repeated steps.

documentation — Clear description of purpose, inputs and expected behaviour.

### New learning

State the precondition — The array must not be empty, or a[0] cannot be read.

Trace the maximum — Set best to 3; compare 7, update best to 7, then ignore 2 and 5.

Document contract — Input: nonempty int array. Output: greatest element. Does not modify input.

### Worked example

Trace Largest([4,2,9]) and write the returned result.

1. Predict first — Start best at first element 4; 2 does not replace it.

2. Model and explain — 9 is larger so best becomes 9; return 9.

## We Do · Practise with guidance

Trace Largest([8,11,6,10]) and state its precondition.

```
best=8; process 11, 6, 10;
```

Predict first

11 replaces initial best 8; neither 6 nor 10 exceeds 11.

Model and explain

The output is 11; the required precondition is a nonempty array.





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Trace Largest([2,5,1]) and state initial and final best.

### Question 2

Write pseudocode to return the minimum of a nonempty array values and explain its precondition.

### Question 3

Evaluate what happens for Largest([-4,-2,-9]) and explain why initialising best=0 would be incorrect.

## Exit Ticket · Answer before checking

Initialising a maximum-tracking variable to 0 is correct for every integer array.